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" A Scientist is not a person who gives
the right answers ; he's the one who
asks the right questions . "

- Claude Levi-Stra

Harnessing The Power of The Atom...

Nuclear energy is a form of energy released from the nucleus , the core of the atom , made up of protons and neutrons . This source of energy can be produced in 2 ways :

1] Nuclear Fusion – when nuclei fuse together.

2] Nuclear Fission – when nuclei of atom split into several parts.

The nuclear energy harnessed around the world to produce electricity is through Nuclear Fission , while technology to generate electricity from Nuclear Fusion is at R&D phase (Research and Development)

Objectives :

1] To learn more about nuclear fission and how electricity is produced from atoms.

2] To understand the mechanism and structure of nuclear reactor.

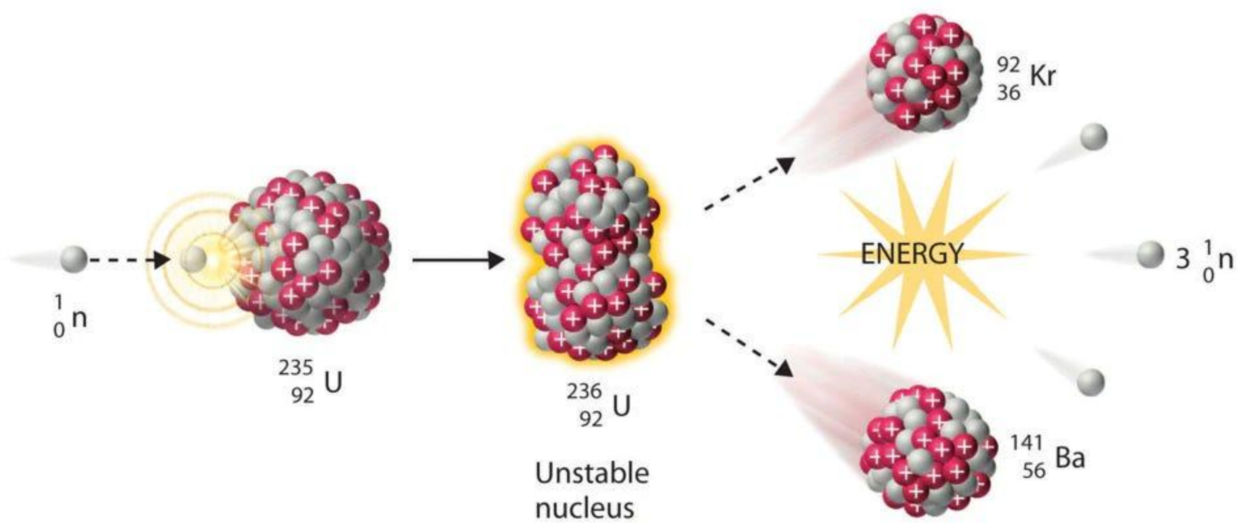
3] To study the effects of nuclear power plants on environment.

4] Learning how nuclear waste is managed .

What is Nuclear Fission :

Nuclear fission is a reaction where the nucleus of an atom splits into 2 or more smaller nuclei while releasing energy . This is a chain reaction which needs to be controlled .

The fission process often produce ‘ Gamma photons ’ (gamma photons are penetrating form of electromagnetic radiation arising from high energy interaction like the radioactive decay of atomic nuclei) and release a very large amount of energy even by the energetic standard of ‘ Radioactive decay ’ (the process by which an unstable atomic nucleus loses energy by radiation) .



A neutron is absorbed by a U-235 nucleus , turning into an excited U-236 nucleus ; With the excitation energy provided by the kinetic energy of the neutron and forces that bind the neutron (strong nuclear forces) . The U-236 splits into fast moving lighter elements i.e.; Kr-92 & Ba-141 also releases several free neutrons (2 or 3) , one or more gamma rays and a large amount of kinetic energy .

Nuclear fission was discovered by chemists “Otto Hahn” & “Fritz Strassmann” and physicists “Lise Meitner” & “Otto Robert Frish”. Hahn and Strassmann predicted the existence and liberation of neutrons during the fission process , opening up the possibility of a nuclear chain reaction.

For heavy nuclides , it is an exothermic reaction which releases large amount of energy , both as electromagnetic radiation and as kinetic energy of the fragments (heating the bulk material where fission take place) . For fission to produce energy the total binding energy of the resulting elements must be greater than that of the starting elements . The fission barrier (the activation energy required to split an atomic nucleus) must also be overcome .

Fissionable nuclides primarily split in interactions with fast neutrons , while fissile nuclides easily split in interaction with slow or thermal neutrons , usually originating from moderation of fast neutrons .

Parameters	Fast Neutrons	Slow Neutrons
Energy level	1-20 MeV	~ 0.025 eV
Speed	~ 20,000 Km/s	~ 2.2 Km/s
Origin	Directly from nuclear fission	When fast neutrons are slowed by a moderator

There are many types of fission :

COLD FISSION : Fission events for which fission fragments have Such low excitation energy that no neutrons or Gamma rays are emitted .

PHOTOFISSION : The process in which a nucleus after absorbing a gamma ray , undergoes nuclear fission and splits into fragments .

How electricity is generated by nuclear reactuion :

Enriched uranium fuel (U-235) pellets are loaded into rods , when a neutron strikes a U-235 nucleus it splits the atom and releases huge amount of heat ; water in a pressurized loop absorbs the extreme heat generated by the chain reaction . This hot water is piped to a steam generator , where it boils a secondary supply of water into high-pressure steam.

The high-pressure steam is directed onto the blades of a massive turbine , causing it to spin at incredible speeds , this converts thermal energy into mechanical energy . The spinning turbine turns a shaft connected to an electrical generator . The generator rotates magnets inside a copper wire coil (solenoid), producing an induced current . After passing through the turbine , the steam is cooled back into water in a condenser so that , the cycle can start over .

Mechanism , Structure and Types of Nuclear Reactor :

A nuclear reactor is a device used to sustain a controlled fission nuclear chain reaction.

They are used for production of commercial electricity , marine propulsion , weapon production & research . Fissile nuclei (primarily U-235 and Pu-239) absorbs single neutron and split , releasing energy and multiple neutrons , which can induce further fission . Reactor stabilizes this , regulating neutron absorbers and moderators in the core . Fuel efficiency is exceptionally high ; low-enriched uranium is 120,000 times more energy-dense than coal .

Heat from a nuclear fission is passed to a working fluid coolant . In commercial reactors , this drives turbines & electrical generator shafts . Some reactors are used for district heating and isotope production for medical and industrial use .

Types of Nuclear Reactors :

1] PWR : Pressurized water reactor ; 277(63.2%)

2] BWR : Boiling water reactor ; 80(18.3%)

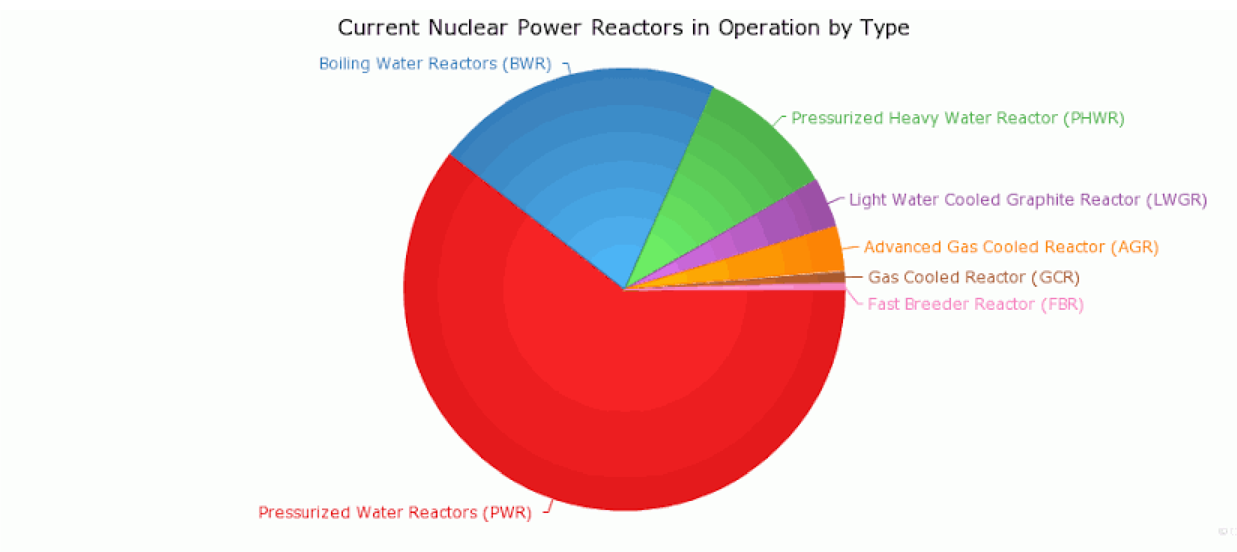
3] PHWR : Pressurized Heavy-water reactor ; 49(11.2%)

4] GCR : Gas-cooled reactor ; 15(3.4%)

5] LGWR : Light water-cooled graphite reactor ;9(2.2%)

6] AGR : Advanced gas-cooled reactor ; 5(1.2%)

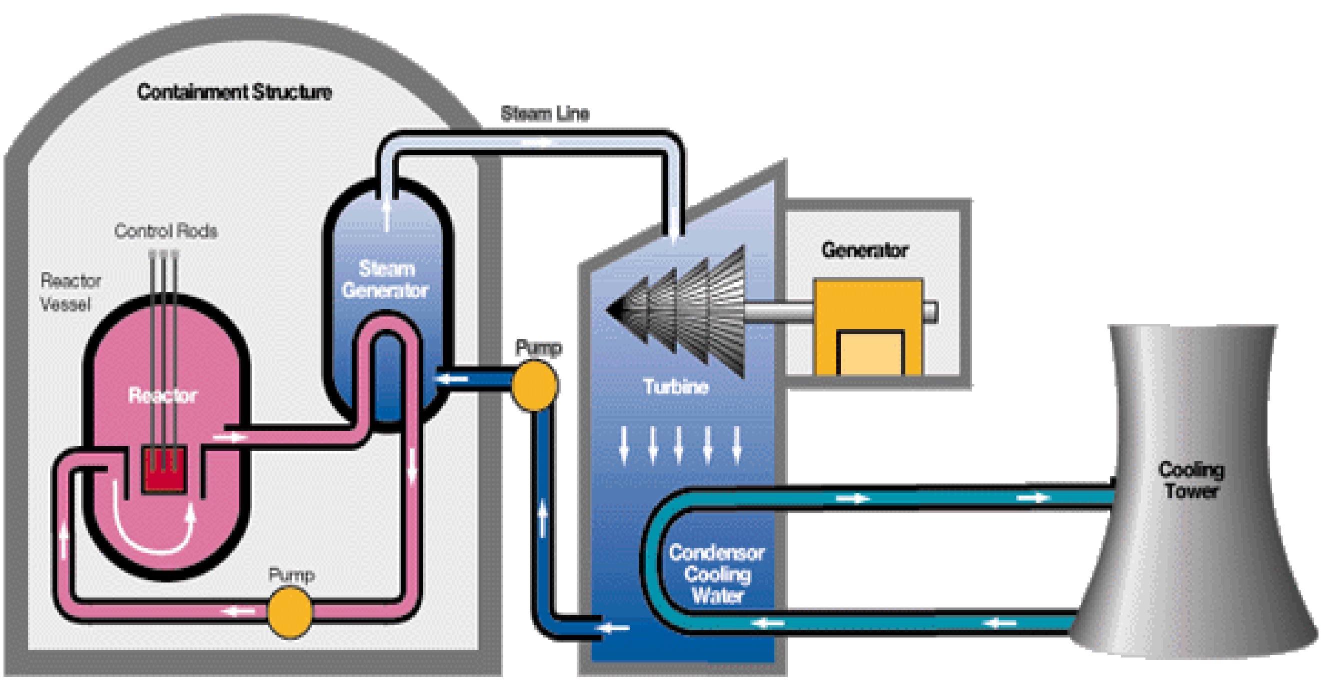
7] FBR : Fast breeder reactor ;2(0.5%)



1] PWR – Pressurized water reactor :

Primary characteristic of PWR is a pressurizer , a specialized pressure vessel . During normal operations , a pressurizer is partially filled with water ; a steam bubble is maintained above it by heating the water with submerged heaters , the pressurizer is connected to the primary reactor pressure vessel (also known as RPV) and the pressurizer bubble provides an expansion space for changes in water volume in the reactor.

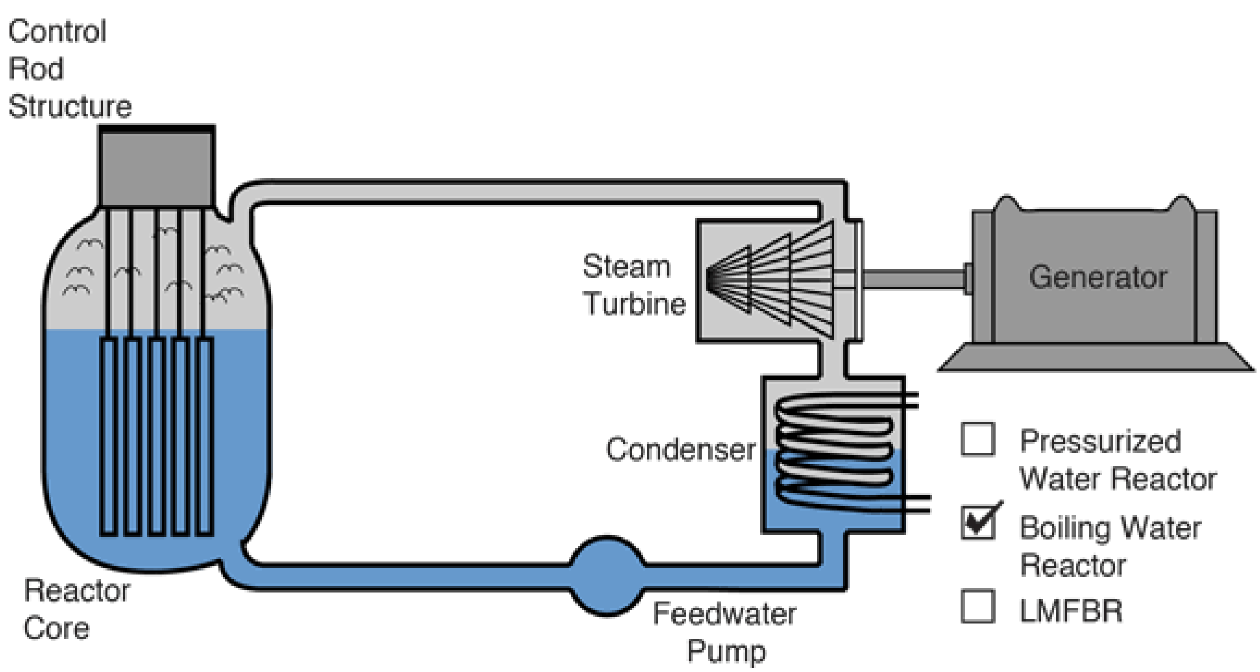
This arrangement also provides a means of pressure control for the reactor by increasing or decreasing the steam pressure in the pressurizer using the pressurizer heaters .



2] BWR – Boiling water reactor :

BWR are characterized by boiling water around the fuel rods in the lower portion of a primary reactor pressure vessel . A boiling water reactor uses U-235 enriched as uranium dioxide, as its fuel . The fuel is assembled in rods housed in a steel vessel that is submerged in water . The nuclear fission causes the water to boil , generating steam . This steam directly flows through pipes into turbines . The turbines are driven by the steam & this process generates electricity .

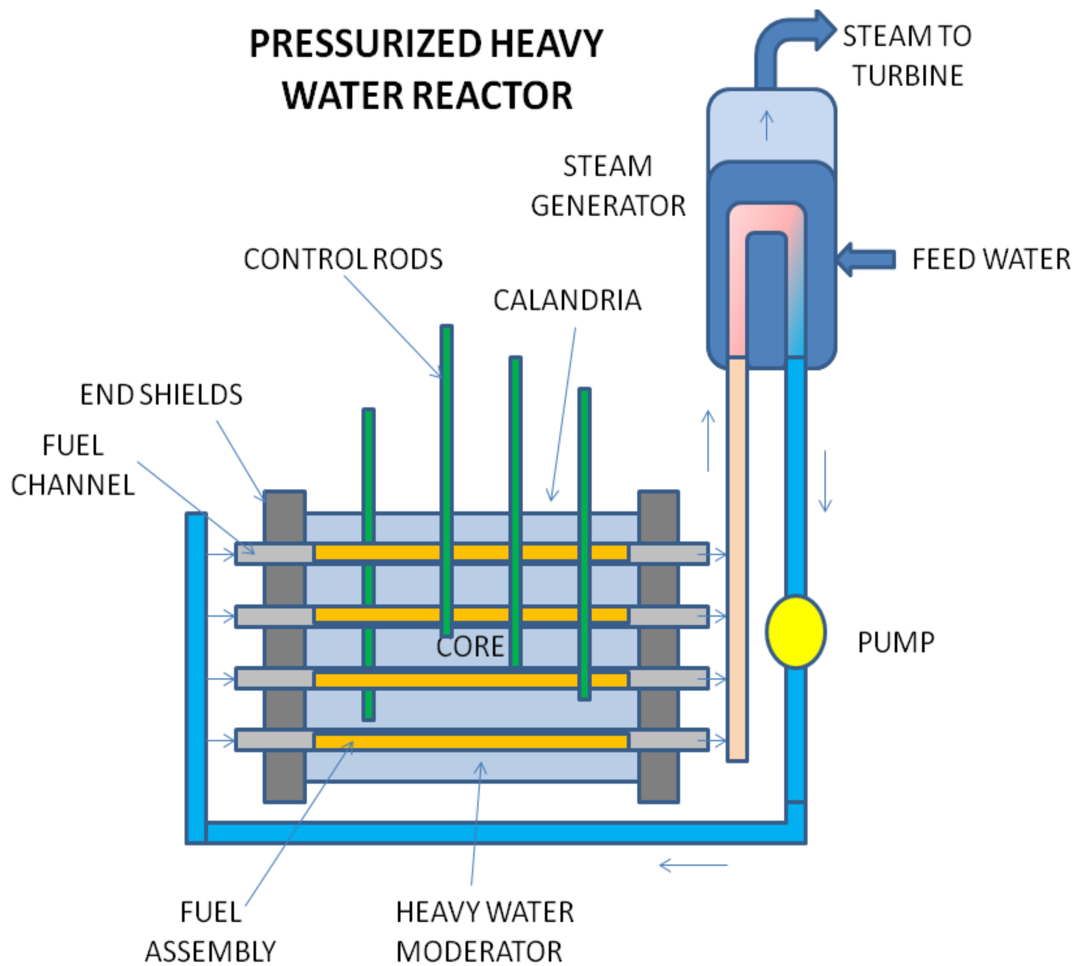
During normal operations , pressure is controlled by the amount of steam flowing from the reactor pressure vessel to the turbine .



3] PHWR – Pressurized heavy water reactor :

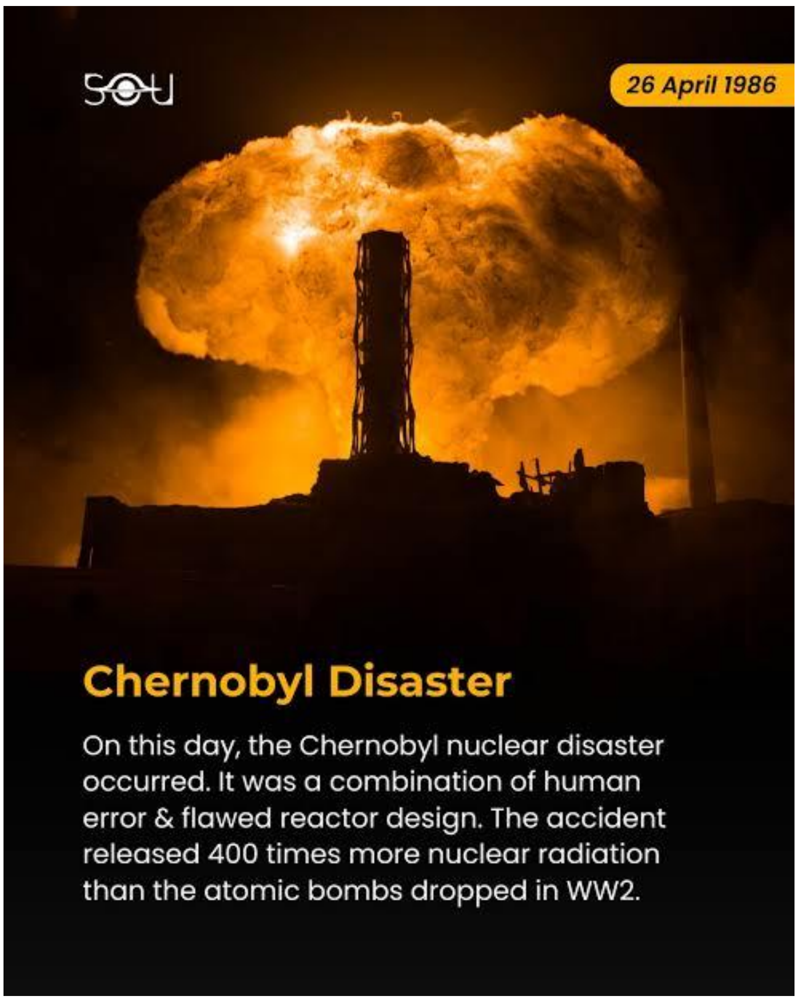
These reactors are subset of pressurized water reactor PWR sharing the use of a pressurized , isolated heat transport loop but using heavy-water as coolant and moderator for the greater neutron economies .

PWR , BWR & PHWR are mostly used reactor types worldwide .



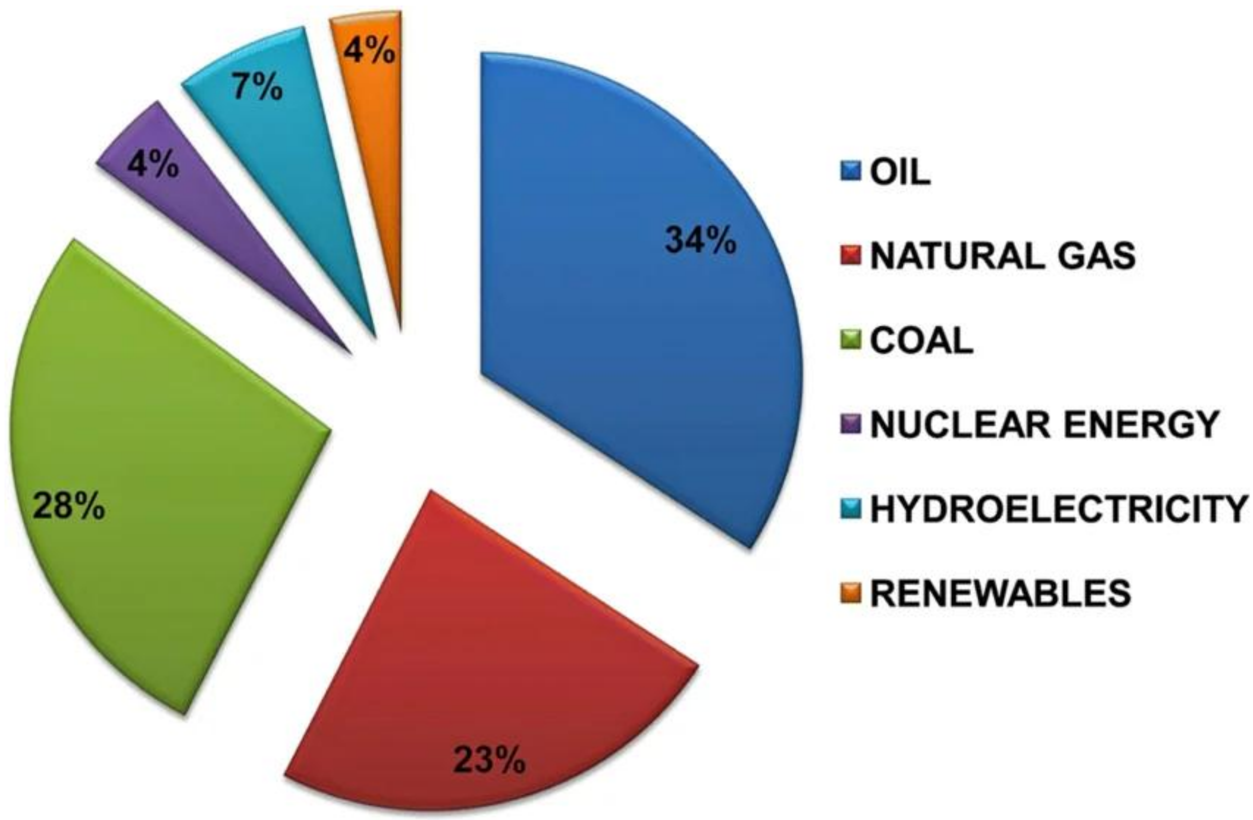
Impact of Nuclear Power Plant on Environment :

Nuclear power has various environmental impacts , both positive and negative . Including the construction and operation of the power plant , the nuclear fuel cycle and the effects of nuclear accidents .



*Chernobyl Disaster is one of the deadliest nuclear disasters

Nuclear power plants do not burn fossil fuels and so do not directly emit carbon dioxide . The carbon dioxide emitted during mining , enrichment of uranium , fabrication and transport of fuel is small when compared with the carbon dioxide emitted by fossil fuels of similar energy yield ; however , these plants still produce other environmentally damaging waste . Nuclear energy and renewable energies have reduced environmental costs by decreasing carbon dioxide emission resulting from energy consumption .



Coal – 820 tones (273 times more than nuclear energy)

Oil – 720 tones (180 times more than nuclear energy)

There is a catastrophic risk potential , if containment fails ; which in nuclear reactors can be brought about by overheated fuels melting and releasing large quantities of fission products into the environment .

In normal operations , nuclear power plant releases less radioactive materials than coal power plants whose fly ash contains significant amount of thorium and its daughter nuclides .

A large power plant may reject waste heat to a natural body of water ; this can result in undesirable increase of the water temperature with adverse effect on aquatic life . Alternative include cooling towers .

Nuclear waste & management :

Radioactive waste is a type of hazardous waste that contains radioactive materials . It is a result of many activities , including nuclear medicine , nuclear research , nuclear power generation etc. The storage and disposal of radioactive waste is regulated by government agencies in order to protect human health & the environment.

Radioactive waste is broadly classified into 3 categories ;

1] Low-level waste (LLW) : such as paper , rugs , tools, clothing which contains small amount of mostly short-lived radioactive substances .

2] Intermediate-level waste (ILW) : contain higher amount of radioactivity and requires proper shielding .

3] High-level waste (HLW) : highly radioactive and hot due to decay heat , thus requires cooling proper shielding

Spent Nuclear fuel can be processed in “Nuclear reprocess plants” one-third of the total amount has already been reprocessed ; with nuclear reprocessing , over 96% of the spent fuel can be recycled back into uranium-based and mixed-oxides (MOX) fuels .

The residual 4% is minor actinides and fission products , the latter of which are a mixture of stable and quickly decaying elements , medium-lived fission products such as Strontium-90 & Caesium-137 and finally 7 long lived fission products with half lives in the range of hundreds of thousands to millions of years.

7 Long-lived fission products :

1] Technetium-99 = 211,000 years half life

2] Tin-126 = 230,000 years half life

3] Selenium-79 = 295,000 years half life

4] Zirconium-93 = 1.53 million years half life

5] Caesium-135 = 2.3 million years half life

6] Palladium-107 = 6.5 million years half life

7] Iodine-129 = 15.7 million years half life

(note : Uranium-235 and is not mentioned above , it also have half live in millions of years .

Uranium-235 = 704 million years half life .)

The minor actinides , meanwhile are heavy elements other than Uranium and Plutonium which are created by neutron capture , their half lives also range in millions of years ; and as a alpha-emitters they are particularly radiotoxic.

The Nuclear waste is subsequently converted into a glass-like ceramic for storage in a deep geological repository.

A future way to reduce waste accumulation is to convert current reactors in favor of generation – IV reactors , which output less waste per power generated .

Fast reactors FBR such as “BN-800” in Russia are also able to consume MOX fuel that is manufactured from recycled spent fuel from traditional reactors .

Associated hazard warning signs :

#Can Nuclear energy realistically replace 50% of India’s coal-based electricity generation by 2050 ? :

India needs an optimal mix of energy resources with a minimum of fossil fuel to meet its rapidly increasing demand for 24/7 clean energy .This demand is driven by economic growth .The government has set a target of a non-fossil energy capacity of 500 GW by 2030 with the main focus on solar and wind energy .

India is the second largest producer and consumer of coal in the world .Coal-based electricity currently meets over 70% of its energy needs . Burning of coal is a leading contribution to air pollution and greenhouse gas emissions , responsible for over 40% of India’s CO_2 emissions . India ranks 3rd in global emission from coal mining .

Nuclear power plants life cycle CO_2 emission per unit of electricity generation is 12 g compared to 820 g in case of coal-based power plants . India has over 60 years of experience in nuclear energy technology . Several organizations like DAE (Department of Atomic Energy) , BARC (Bhabha Atomic Research Center) , Atomic Energy Regulatory Board , etc. are contributing to this .

India is also a member of the 35-membered collaborative effort project the " International Thermonuclear Experimental Reactor " for advancing magnetic fission .

Coal-based power plants (in India) :

- 285-300 power plants
- 62% of country's energy need
- Total operational capacity exceeds 205,000 MW
- Largest plant – Vindhyachal Super Thermal Power Station in Madhya Pradesh having capacity of 4760 MW

Nuclear power plants (in India) :

- 7 operational plants
- 24 functional reactors
- Total operational capacity of 8780 MW
- Long-term goal – at least 100 GW of Nuclear energy capacity by 2047

$$[\ E = nP\ .\ t\ ]$$

E = Energy required per hour

n = no. of nuclear power plants ; t = time (hr)

P = capacity of power plant

India’s avg. need of electricity per hour (E) – 250 GWh

Avg. capacity of nuclear power plants in India – 1268 MW

(1 GW = 1000 MW)

Total nuclear power plants in India – 7

62% of total power is generated by coal ,

Therefore, power generated by coal = 250,000 * 62 /100

= 155,000

If we wish to replace 50% of coal-based power plants by 2050 , then ;

155,000 * 50/100 = 77500

77500 MW of power must be generated by Nuclear power plants .

[E = nP.t]

77,500 = n(1268).3 .... (if we want t =3 hrs)

n = 77,500/(1268.3)

= 77,500/3804

= 20.373 ~ 20

Current nuclear power plants – 7

Required nuclear power plants – 2

Avg. cost to build one nuclear power plant = 8.8 billion$

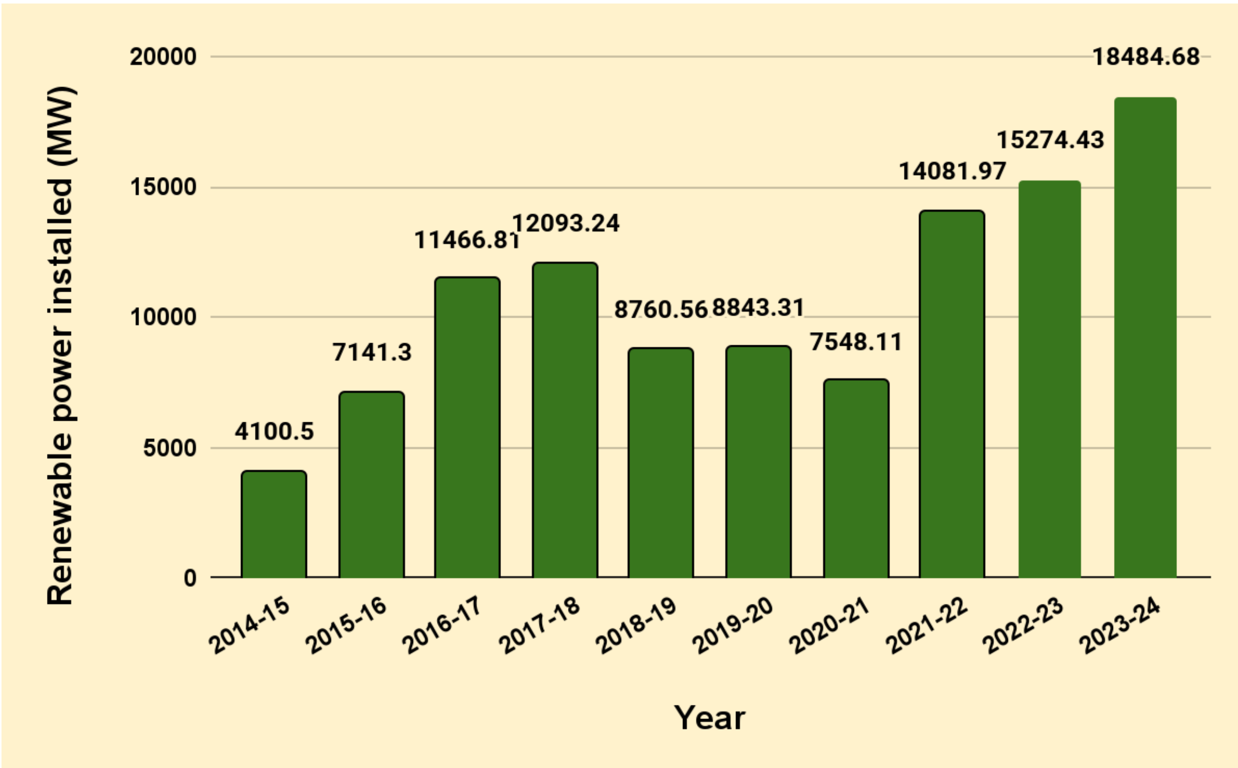
Required funds to build more 13 power plants = 13*$8.8 billion

= $114.4 billion

Indian Government's total investment in Nuclear technologies by 2047 is expected over $241 billion .

Conclusion : To replace 50% of coal-based power plants by Nuclear power plants by 2050 would require construction of more 13(approximately) nuclear power plants ; which will take approx 14-16 years and a amount of $114.4 billion .

We can say that ; Yes , Nuclear energy can realistically replace 50% of India's coal-based electricity generation by 2050



(Graph of India's electricity demand over years)

Nuclear Power and India :

1] BARC :

The Bhabha Atomic Research Center is India's premier nuclear research facility , headquartered in Trombay , Mumbi , Maharashtra , India . It was found by Dr. Homi Jehangir Bhabha as the "Atomic Energy Establishment , Trombay" (AEET) in Jan 1954 .

It operates under Department of Atomic Energy (DAE) , BARCs core mandate is to sustain peaceful applications of nuclear energy . It manages all facets of nuclear power generation ; from the theoretical design of reactors to computer modeling and simulation , risk analysis , development and testing of new reactor fuels . BARC operates a number of research reactors across the country to test new reactor fuels .

2] India's three-stage programme :

India's 3-stage Nuclear programme was formulated by Homi-Bhabha in the 1950s , to secure the country's long term energy independence , through the use of uranium and thorium reserves found in the monazite sands of coastal regions of south India .

The ultimate focus of the programme is on enabling the thorium reserves of India to be utilized in meeting the country's energy requirements . Thorium is particularly attractive for India , as it has only around 1%-2% , but one of the largest shares of global thorium reserves at about 25% of the world's known Thorium reserves .

- ## Stage I : PHWR

In the 1st stage of the programme , natural uranium fueled PHWRs produce electricity while generating plutonium-239 as by product . PHWRs was a natural choice for implementing the 1st stage , because it had the most efficient reactor design in terms of uranium utilization and existing Indian infrastructure in the 1960s allowed for quick adoption of the PHWR technology . Natural uranium contains only 0.7% of the fissile isotope U-235 , most of the remaining 99.3% is U-238 which is not fissile ; but can be converted in a reactor to the fissile isotope plutonium-239 . Heavy water is used as moderator and coolant .

- ## Stage II : FBR

In the 2nd stage , FBRs would use a mixed oxide MOX fuel made from plutonium-239 , recovered by reprocessing spent fuel from the 1st stage and natural uranium . In FBRs , plutonium-239 undergoes fission to produce energy , while the uranium-238 present in the MOX fuel transmutes to additional plutonium-239 . Thus , the stage II FBRs are designed to breed more fuel than they consume . Once the inventory of plutonium-239 is built up , thorium can be introduced as a blanket material in the reactor and transmuted to uranium-233 for use in the 3rd stage , the surplus plutonium breed in each fast reactor can be used to set up more such reactors and might thus grow the Indian civil nuclear power capacity till the point where the 3rd stage reactors using thorium as fuel can be brought online . The design of the country's 1st fast breeder , called Prototype Fast Breeder Reactor (PFBR) , was done by "Indira Gandhi Center for Atomic Research" (IGCAR)

- ## Stage III : Thorium based reactors

A stage III reactor or an advanced nuclear power system involves a self-sustaining series of thorium-232-uranium-233 fuelled reactors . This would be a thermal breeder reactor , which in principle can be refueled after its initial fuel charge using only occurring thorium . According to the 3-stage programme , Indian nuclear energy could grow to about 10 GW through PHWRs fueled by domestic uranium and the growth above that come from FBRs till about 50 GW . The 3rd stage is to be deployed only after this capacity has been achieved .

3] Thorium Reserves :

India holds the largest Thorium reserves in the world , accounting for roughly 25% of global deposits . The country's in-situ reserves are estimated at approximately ~ 1.07 million tones of thorium contained within ~ 11.93 million tones of monazite sand .

The majority of these deposits are found as monazite placer sand along coastal areas and in river beds . The leading states for these reserves are :

1.Andhra Pradesh

2.Tamil Nadu (Kanyakumari-Manavalakurichi belt)

3.Odisha

4.Kerala

Current Technologies :

1] Small Modular Reactors (SMR) :

A SMR is an emergent class of nuclear fission reactors with a rated electrical power of less than 300 MW , which use modular design principles to achieve streamlined construction & enhanced scalability compared to large light water reactors (LWR) . Many SMR designs are intended to be built in factories and transported to installation sites as prefabricated modules , while others are designed for flexible multi-unit configurations .

The term SMR refers to the physical size , electrical capacity and modular construction approach . Reactor technology varies significantly among SMR designs , As of march 2026 most of SMR designs are light-water reactors . Commercial SMRs are designed to deliver an electrical power output as low as 10 MW and up to 300 MW per module . Reactors below 10 MW in capacity are considered as “Nuclear micro reactors”. SMR may also be designed purely for process heat rather than electricity . Many SMR designs plan for customers to simply add modules to achieve a desired electrical output rather than scaling the size of the reactor . These reactor are also expected to enhance safety through Passive Safety systems that operate without external power or human interference .

2] Molten Salt Reactor :

A molten salt reactor (MSR) is a class of nuclear fission rector in which the primary nuclear reactor coolant and the fuel is a mixture of molten salt with a fissile material .

Two research MSRs operated in United States in the mid - 20th century . The 1950s “Aircraft Reactor Experiment” (ARE) was primarily motivated by the technology’s compact size , while the 1960s “Molten-Salt Reactor Experiment” (MSRE) aimed to demonstrate a nuclear power plant using a thorium fuel cycle in a “Breeder Reactor” .

3] Generation-IV Reactor :

Increased research into generation IV reactor designs renewed interest in the 21st century with multiple nations starting projects . On October 11,2023 China’s “TMSR-LV 1” reached criticality &subsequently achieved full power operation as well as thorium breeding .

Cost Analysis :

	Nuclear energy	Solar energy	Wind energy	Coal-based energy
Cost per KWh	3.5 – 4.5 rupees	2.3 – 3 rupees	2.5 -3.5 rupees	3.5 – 5 rupees
Life span	40 – 80 yrs	25 – 30 yrs	20 – 25 yrs	35 – 40 yrs
Reliability	>90%	20% -25%	30% -40%	40% - 50%
Land use	10 – 12 hectares	1000-2000 hectares	2000-3000 hectares	4000-5000 hectares

By comparing this , although the initial cost for Nuclear energy being high ; it is the most efficient and reliable source of energy.

Reactor Design Proposal :

Concept : Advanced Fast Breeder Thorium Reactor (AFBTR) .

Objectives : 1] Reactor that uses thorium as the long term fuel source .

2] Breeds more fissile fuel than it consumes.

3] Produces less long-lived nuclear waste

Core fuel cycle :

Initial startup fuel – plutonium-239 or uranium-233

Blanket region – Thorium-232

Breeding process :

Thorium-232 + n → Thorium-233

Thorium-233 → Protactinium-233 (by beta decay)

Protactinium-233 → Uranium-233

The resulting Uranium-233 becomes reactor fuel for further nuclear chain reactions .

Passive Safety System :

1] Gravity-driven emergency cooling –

If power is lost ;

- Valves automatically open
- coolant flows downward by gravity
- No external electricity required .

2] Molten salt backup drain tank -

(inspired by MSRs) If temperature becomes too high ;

- A freeze plug melts
- Fuel drains into a safe underground tank
- Chain reaction stops automatically

Waste reduction :

AFBTR can ;

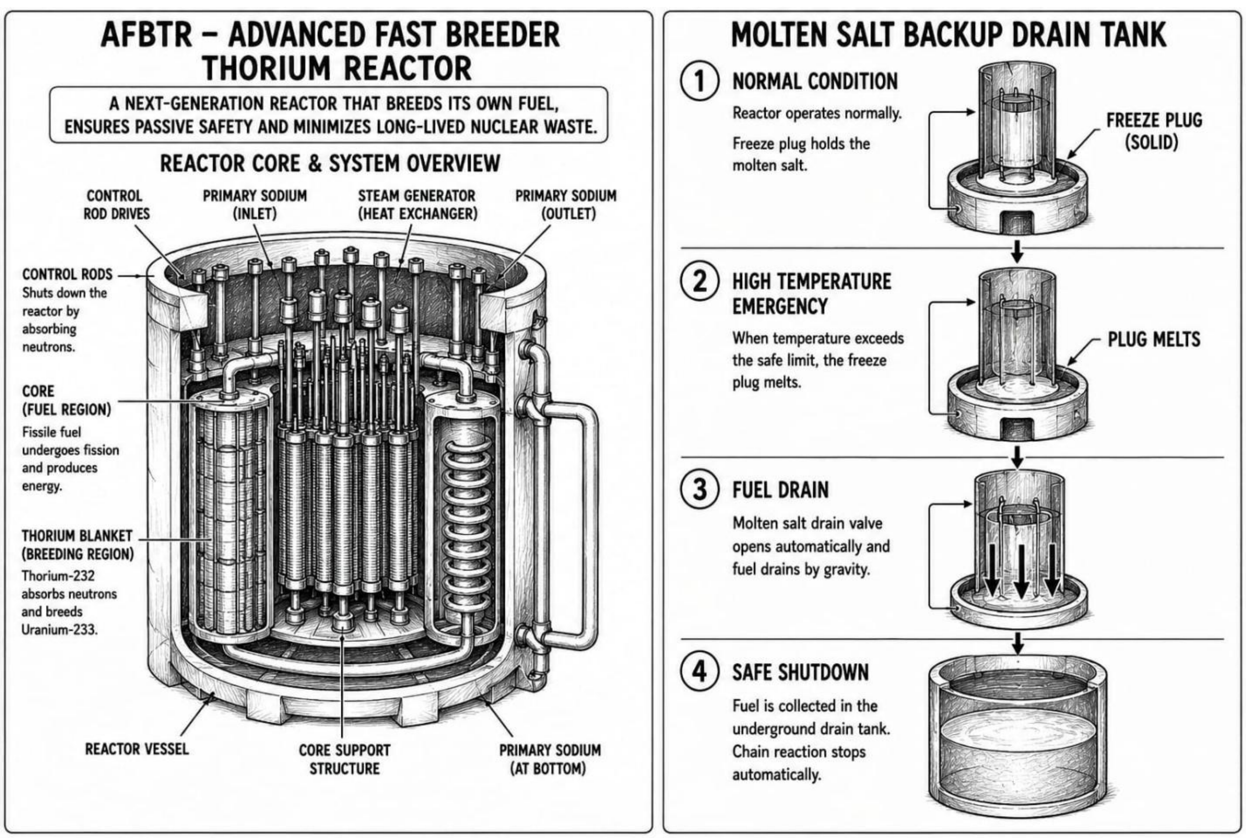
- Recycle spent fuel into MOX fuel
- Burn minor actinides
- Reduce long-lived waste inventory

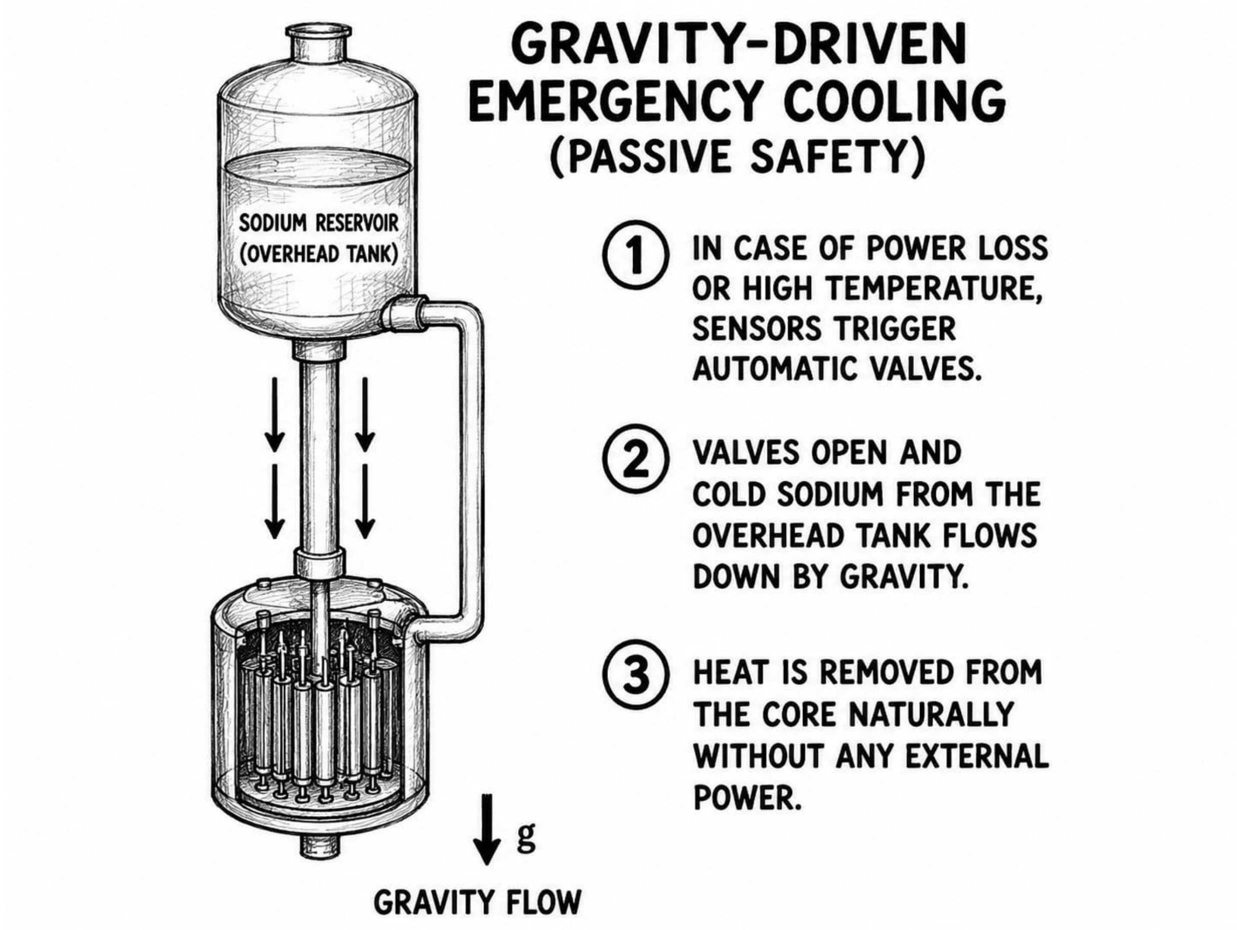
Indian Application :

India has largest thorium reserves around ~ 1.07 million tones of thorium ; making it a 25 % share holder in thorium reserves . With the help of AFBTR India's need for domestic uranium will decrease , resulting in less uranium imports ; which will help in strengthening India's GDP . With AFBTR India will get long-term energy independence .

Parameters	Proposed values
Reactor type	FBR
Fuel	Thorium-232 & Uranium-233
Coolant	Liquid sodium
Electrical output	~ 500 MW
Breeding ratio	>1.0
Safety system	Passive safety system
Waste Reduction	Very-high

Model of AFBTR with Molten-salt backup drain tank and Gravity driven emergency cooling system (passive safety system)





NOTE :

- This concept requires advanced fuel reprocessing
- Has high initial construction cost
- The thorium fuel cycle is still under development
- Engineering challenges

(these all are limitations of this prototype)

Limitations of My Study :

This study is based on publicly available data & simplified engineering assumptions . Detailed reactor economics and national energy policy considerations were beyond the scope of this project .

Future Work :

Future work could include neutron transport modeling , economic optimization studies or detailed prototype model of AFBTR .

Simulation of Nuclear Reactor :

https://atom-chain-reaction.preview.emergentagent.com/

Conclusion :

- Nuclear energy is sustainable and has most effective fuel to energy ratio .
- Instead of only Nuclear energy combination of all sustainable energies like solar , wind , hydro etc. can result in future of energy generation .
- There are some problems with Nuclear , energy , but by the time due to strict safety majors , passive safety systems and advanced technology it has reduced to just 2% - 3% .
- To make Nuclear power more efficient , the power plants should be upgraded to Generation – IV reactor , so that they can consume the MOX fuel ; which will result in ~ 99% - 100 % efficiency by using the spent fuel .
- In future advanced reactor designs like AFBTR can improve the efficiency of reactors , help in developing and testing of thorium mixed fuels .

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